

BASF

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Oximony®

Longevity Hero for skin that defies time

Global Except US

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SUMMARY FILE

Oximony®	BC10187
Origin - Description	Natural extract of *Lysimachia christinae* aerial parts, from a FairWild certified sourcing
Regulatory data	
INCI	Maltodextrin (and) Lysimachia Christinae Flower/Leaf/Stem Extract
China	All components are listed in Inventory of Existing Cosmetic Ingredient in China (IECIC 2021)
CAS# ENEICS	9050-36-6; 3036677-64-9 232-940-4; /
Appearance	Fine powder, beige to light brown, with weak characteristic odor
Natural labels	Cosmos approved pending at the time of printing
Naturalness content (ISO 16128)	100% from Natural Origin
Cosmetic use	
Properties	• Helps to increase skin cell longevity by combating senescence and increase in cellular content of Taurine and its transporter • Helps to protect skin cells against oxidative stress • Helps to decrease skin fatigue • Contributes to improve skin firmness • Helps to increase skin glow
Applications	Face care, anti-fatigue, anti-aging and glow perfection
Formulation data	
Concentration of use	0.25%
Solubility	Soluble in water, insoluble in oils and fats
Incorporation method	Incorporate into the cosmetic product up to 80°C, or at room temperature for cold processing
Optimal pH	4-7



Longevity: elevating beauty beyond age

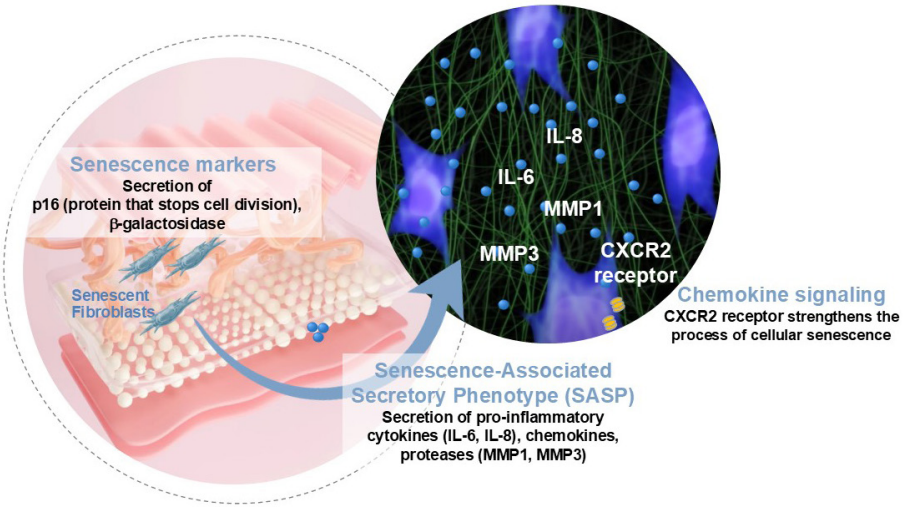
In a world where life expectancy continues to rise, beauty must evolve to meet the needs of consumers who want to live long, healthy lives. The global population is aging, with 22% expected to be over 60 by 2050 [SilverEco, 2009]. The desire to age healthy has become a deep aspiration. Aging is no longer an enemy to fight but a process to embrace. Beauty now focuses on extending the lifespan of the skin, by shifting to combating visible signs of aging toward a more holistic approach centered around rebuilding skin health and vitality at every level [Wyles *et al*, 2024]. Our relationship with aging is evolving, reflected by the shift in cosmetic products towards longevity beauty, replacing the old anti-aging narrative of the 2000s [Sykes, 2023]. Today, it's about supporting long-term skin health, vitality, and glow to recover youthful skin allowing us to age gracefully and powerfully.

The expanded hallmarks of aging encompass a myriad of cellular processes including genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, deregulated nutrient-sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, altered intercellular communication [López-Otín *et al*, 2023]. These interconnected mechanisms can accelerate aging if intensified. Among these hallmarks, cellular senescence emerges as a linchpin, a phenomenon intricately linked to the root-cause of aging process [Wyles *et al*, 2024].

Fighting cellular senescence is key to keep skin vitality [Marcheggiani *et al*, 2021], however senescence just represent one of the hallmarks of aging even if it's a major one. To sustain vibrant, youthful healthy skin, a comprehensive approach that targets hallmarks such as senescence can be effective.

Oximony has been developed by BASF BCS to set up a new skincare with a concept that combines longevity, and a commitment to sustainability through a FairWild certification of the plant harvest.

Senescence: a main feature of aging



Excessive accumulation of senescent cells in skin can negatively affect its longevity and regenerative capacities [Franco *et al*, 2022]. Cellular senescence is an irreversible proliferative arrest in damaged normal cells in which there is a stop in the cell cycle and namely in cell division [Roger *et al*, 2021]. Senescent cells present unique morphological and molecular characteristics as well as functions that distinguish them from other nondividing cell populations, such as quiescent cells and terminally differentiated cells [Roger *et al*, 2021].

Cellular senescence results in morphological alterations. Senescent cells are enlarged and have an irregular shape; their nuclear integrity is compromised due to the loss of lamin B1, which also leads to the appearance of cytoplasmic chromatin fragments (CCFs); they have an increased lysosomal number and content, which is often detected as high beta-galactosidase activity; they have large but dysfunctional mitochondria that produce high levels of reactive oxygen species (ROS) [Hernandez Segura *et al*, 2018] (Fig. 1).

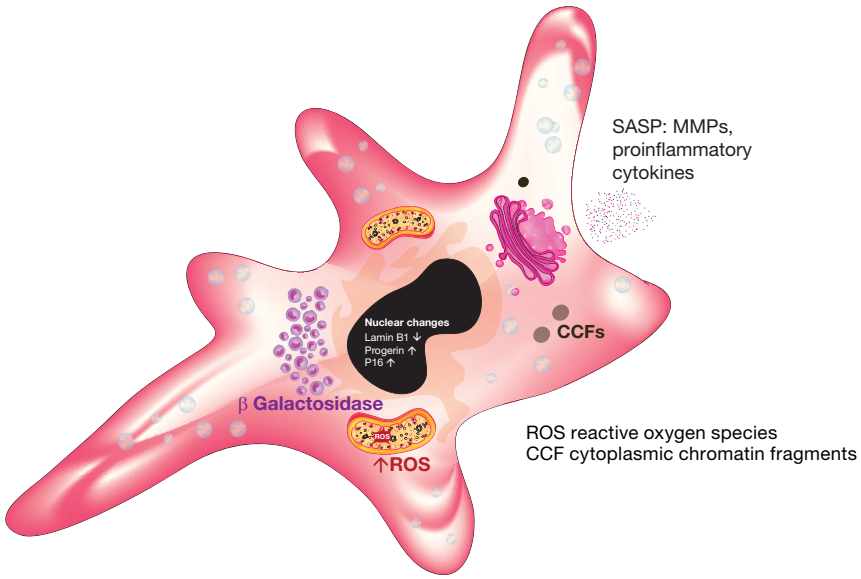


Figure 1 - Morphological and functional alterations of senescent cells [Adapted from Hernandez-Segura *et al*, 2018].

From functional point of view, senescent cells present prolonged cell cycle arrest, correlated with expression of markers like the protein p16 also known as CDKN2A or cyclin-dependent kinase inhibitor 2A and p21 also known as cyclin-dependent kinase inhibitor 1 or CDK-interacting protein 1, both leading to cell cycle arrest. Senescent cells possess a bioactive secretome, known as the senescence-associated secretory phenotype (SASP), exhibit macromolecular damage, and deregulated metabolism [Roger *et al*, 2021].

There are two types of senescence:

Replicative senescence was the **first cellular senescence** subtype to be described. It refers to the decrease in proliferation potential observed after multiple cell divisions that ultimately leads to total cell division arrest. Replicative senescence has been shown for the first time *in vitro* by Hayflick [Hayflick *et al*, 1961]. It can be characterized by an increase in p16 protein, which has been shown to increase during accumulation of senescent cells *in vivo* during aging [He *et al*, 2017]. The shortening of telomeres, as a consequence of multiple cell divisions, has been reported to be causative for this type of senescence [Sharpless *et al*, 2015].

The **second type of senescence** is the stress-induced cellular senescence triggered by extrinsic factors like UV irradiation, pollution and cigarette smoke [Dorf *et al*, 2024]. These stresses cause ROS accumulation generated by mitochondria [Cavinato *et al*, 2024] or directly by UV which in turn cause DNA damages, mitochondria dysfunction [Gu *et al*, 2022] and ultimately senescence. ROS can also act directly as an intracellular second messenger, regulating intracellular signaling pathways and inducing the phenomenon of cellular senescence [Zhu *et al*, 2024].

Histone deacetylase 2 (HDAC2) is an enzyme that regulates various cellular processes. HDAC2 has been shown to protect against DNA damage response and cellular senescence, premature aging via an epigenetic mechanism in response to oxidative stress [Yao *et al*, 2012]. Thus, chemicals able to inhibit HDAC2 like Sodium Butyrate can induce senescence [Xiao *et al*, 1997].

Biomarkers of cellular senescence

Beta-galactosidase (β-gal) is a lysosomal enzyme that is expressed in all cells and that has an optimum pH activity at 4.0. However, during senescence, lysosomes increase in number, size and, consequently, senescent cells accumulate β-galactosidase. The increased amounts of this enzyme make detection of its activity possible even at a suboptimal pH 6.0. The expression of β-galactosidase in skin cells has long been identified both *in vitro* and *in vivo*. This marker is one of the most widely used markers of senescent cells [Franco *et al*, 2022; Wang *et al*, 2024].

Cellular senescence is characterized by a stable cycle arrest established by the activation of some pathways like inhibition of cyclin dependent kinases (CDKs). CDKs phosphorylate and regulate multiple proteins involved in cell cycle progression. Main drivers of the cell cycle arrest in senescence are the CDK inhibitors such as p16INK4a, (hereafter p16), CDKN2B (p15INK4b, hereafter p15) and CDKN1A (p21CIP, hereafter p21) [Roger *et al*, 2021; Wang *et al*, 2024]. p16 is a 16 KDa protein that directly interacts and inhibits CDK4/6. p16 is often used as a unique and specific marker for senescence and its activation has been used extensively to report the presence of senescent cells *in vivo* [Roger *et al*, 2021].

Senescence Associated Secretory Phenotype (SASP) accelerates the skin aging

Senescent cells secrete different chemokines, receptor to cytokine (CXCR2), pro-inflammatory cytokines (IL-8; IL-6), growth factors, and matrix remodeling enzymes (MMP1, MMP3) that define the SASPs. This dynamic and complex secretory activity is one of the main features of senescent cells [Roger *et al*, 2021; Tchin *et al*, 2023]. MMP1 and MMP3 degrade extracellular matrix impairing skin biomechanical properties observed on aged skin. Proinflammatory cytokines generate inflammation which participate in skin aging and accelerate it.

Some SASP factors that strongly accumulate in the senescent cells, when secreted spread and reinforce the senescence signaling in an autocrine fashion [Acosta *et al*, 2013]. Specifically, the cytokine IL-8, its receptor, CXCR2 and other ligands strengthen growth arrest in senescent cells by activating a self-amplifying secretory loop through their ability to boost DNA damage response signaling [Roger *et al*, 2021]. It has been shown that inhibition of CXCR2 expression results in a significant decrease in senescent cells, indicating that CXCR2 could regulate cellular senescence [Zhu *et al*, 2024].

Senescence mitigation for skin longevity

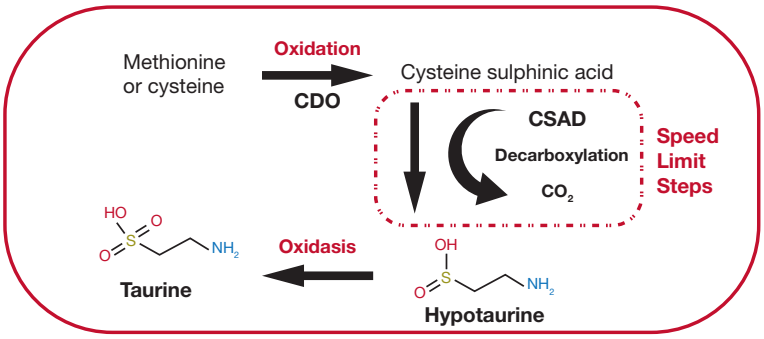
It has been reported that there is an inverse relationship between senescence cell accumulation and longevity [Kumar *et al*, 2023]. There are several options that lead to the decrease of senescence level or its effects: It's possible to use senolytic compound that eliminate senescent cells, to inhibit senescence development or to use compounds with senostatic effect which inhibit SASP or their effect [Sun *et al*, 2022].

Finally, stress induced cellular senescence may be fight by reducing the overall level of oxidative stress. The level of oxidative stress can be reversed in part by proteins expressed following DNA Damage, particularly those implicated in autophagy regulation, which is one of the escape mechanisms from senescence.

Taurine: the secret of skin longevity

Taurine, a naturally occurring amino acid in the body, is one of the most abundant amino acids found in the organisms across eucaryota phyla. In mammalian cells, taurine is produced from cysteine by the action of an enzyme, the cysteine sulphinic acid decarboxylase. Taurine can also be obtained from diet and is taken up by cells through taurine transporter [Lambert *et al*, 2015] (Fig. 2). Taurine is also used in energy drinks as it is involved in energy metabolism of muscles [Wen *et al*, 2019].

At the cellular level, taurine participates in cell homeostasis. It is involved in cells osmoregulation [Siefken *et al*, 2003] and protein folding [Ito *et al*, 2024]; these effects are possible only if its transporter TauT is expressed, as taurine cannot enter passively in cells [Siefken *et al*, 2003; Ito *et al*, 2024]. Moreover, it has been reported that taurine accumulation can protect keratinocytes from UVB-induced apoptosis, by regulating the cell osmosis through TauT [Janeke *et al*, 2003].



CDO: Cysteine DiOxygenase - **CSAD**: Cysteine Sulphinic Acid Decarboxylase

Figure 2 - Taurine synthesis.

Taurine in its native form and when transformed in taurine chloramine exerts strong antioxidant and anti-inflammatory activity; it protects against UVB-induced skin inflammation by promoting the expression of Nrf2 mediated antioxidant / anti-inflammatory enzyme and suppress pro-inflammatory gene expression [Kim *et al*, 2021].

Taurine deficiency may contribute to aging, with supplementation showing promise in improving longevity and health in mammals. Taurine molecule is able to target nearly all hallmarks of aging: Genomic instability, Telomere attrition, Epigenetic alteration, Loss of proteostasis, Nutrient-sensing, Mitochondrial dysfunction, Stem cell exhaustion, Intercellular communication, Inflammation and cellular senescence [Singh *et al*, 2023] (Fig. 3).

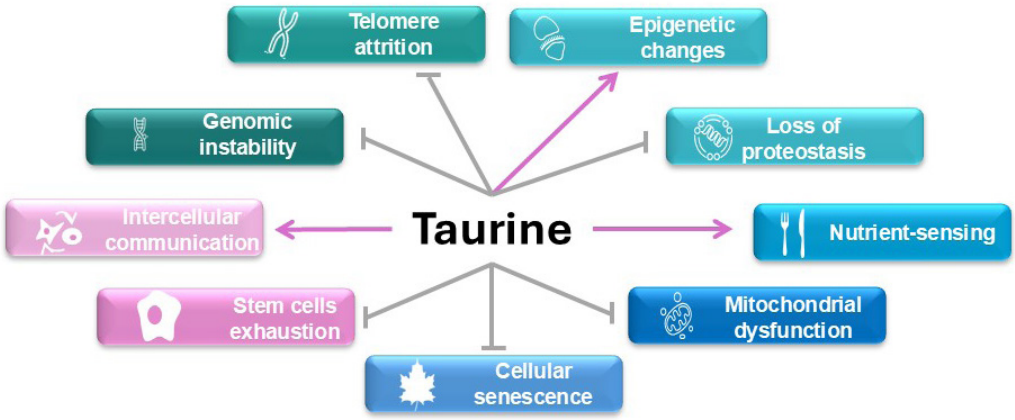


Figure 3 - Taurine effects on key hallmarks of aging: ⊣ inhibition - → promotion.


Lysimachia christinae

A sustainably sourced plant for skin longevity improvement

Oximony is a *Lysimachia christinae* extract characterized by its content in polyphenols with a fully traceable and FairWild certified sustainable sourcing.

Description of the plant

Family: *Primulaceae*

Botanical name: *Lysimachia christinae*

Vernacular name: Guoluhuang, Christina Loosestrife,

TCM name: Jin-Qian-Cao (金钱草)

Synonym: Lysimachia fargesii

Description: *Lysimachia christinae* is a creeping, perennial herbaceous plant that can grow from 20 to 60 cm. It forms mats by developing its network of roots and creeping stems. It is a fast-growing climbing vine. The leaves are green, oval to orbicular or even kidney-shaped, glabrous and shiny, alternate with petioles 1 to 3 cm long. The stems are delicate, grayish-green or reddish-purple in color. The flowers are yellow, solitary, campanulate, with 5 oval to lanceolate lobes, with 5 stamens [Hu *et al*,1996].
The plant is hermaphrodite and is pollinated by insects. It flowers from May to June, and gives fruit from July to October.

Distribution: *Lysimachia christinae* grows up to an altitude of 400 to 2300 m; It grows in sloping hills, and in wetlands at the edge of and inside forests.

Habitat: damp areas along streamsides, open forests, forest margins; at 500-2300 m [Hu *et al*, 1996].

Origin: China

Part used: aerial part (leaves, stems, flowers).

Lysimachia christinae: traditional and modern uses

In traditional Chinese medicine, Jin Qián Cão (*Lysimachia christinae*) acts on Qi by eliminating humidity-heat and promoting energy circulation in the liver and gallbladder [www.passeportsante.net].

The juice of the fresh whole plants of *Lysimachia christinae* (*L. christinae)* mixed with rice washing water or alcohol is orally used to treat furunculoses [Editorial Board of 'Zhonghua Bencao']. The juice of the fresh flowers and leaves of *L. christinae* mixed with lime and tung oil is externally used to treat burns and scalds [Editorial Board of 'Zhonghua Bencao']. The juice of the fresh whole plants of *L. christinae* is orally used and their residues are externally used to treat snake bites [Editorial Board of 'Zhonghua Bencao']. The mixed decoction of the whole plants of *L. christinae* and the roots of Arnebia euchroma (Royle) Johnst. was externally used to treat keloids [Wang RY *et al*, 1990].
Regarding general health, it has been also reported diuretic, detumescent, and detoxifying effects for *Lysimachia christinae* [Wu *et al*, 2018].

SUSTAINABLE SOURCING following FairWild certification guidelines

Oximony embodies a commitment to sustainability and ethical sourcing.

Lysimachia christinae aerial parts (leaves, stems, flowers) are wild harvested in China (Sichuan province).

As part of our Responsibly Active program, this sourcing is certified by FairWild, following 7 principles and 24 criteria, audited annually by an independent certification body, promoting sustainable harvesting and fair trade.

FairWild certification protects plant resources (no negative environmental impacts), respects local rights, ensures fair trade, and supports worker rights.

Additional information on *Lysimachia christinae* aerial parts sourcing are available in the Botanical Sourcing information (available upon request).



FairWild certification

The FairWild Standard has several principles and criteria addressing three themes: the ecological, socio-cultural, and business requirements for sustainable wild collection.

Ecological requirements: wild harvesting supports conservation of the collected species. Wild harvesting has a neutral or positive impact on the collection area and the broader landscape.

Socio-cultural requirements: human rights of wild collectors and workers are respected. Arrangements between operators, wild collectors, and workers are fair. Benefits and respect for customary rights of wild collectors and their communities are ensured.

Business requirements: harvest and trade of wild resources are legally compliant. Responsible business practices are applied.

As part of the *Lysimachia christinae* harvest, the FairWild certification guarantees:

- **Implementation of a resource management plan** to ensure sustainable practices.
- **Adherence to strict collection guidelines**, including clearly defined harvesting techniques and limits on quantities.
- **Preservation of species populations**, ensuring that harvesting does not negatively impact their sustainability or survival.
- **Protection of biodiversity and ecosystems**, ensuring that other species, habitats, and the broader landscape remain unaffected.

From a social perspective, the certification also confirms:

- **Strong partnerships with local stakeholders**, built on principles of traceability and fairness.
- **Long-term contracts for team members**, providing stability and security.
- **Fair compensation**, including income above the local minimum wage
- **Distribution of a fair premium**, ensuring additional benefits for the local community.

Processing and composition



Figure 4 - Processing and composition.

Oximony is obtained through a nature-friendly extraction process using water as the sole solvent. Following a first step of aqueous extraction from the dried and crushed aerial parts, the unextracted insoluble matter is removed and the product is concentrated to obtain Oximony as a spray-dried preservative-free powder with maltodextrin (10-30% Lysimachia Christinae Flower/Leaf/Stem Extract; 70-90% Maltodextrin), characterized by a content in polyphenols ≥ 0.10% (Fig. 4).

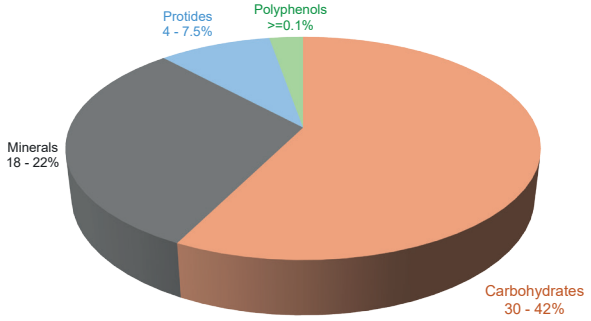


Figure 5 - Typical phytochemical composition of Lysimachia Christinae Flower/Leaf/Stem Extract- in Oximony (weight % on dry matter, based on 4 representative batches).

Typical composition of Oximony active matter is presented in Figure 5. The main components extracted from *L. christinae* aerial parts are carbohydrates, protides and polyphenols which have been quantified respectively by total sugars analysis using colorimetric phenol method, by measurement of total nitrogen by Kjeldhal method (conversion factor of 6.25) and colorimetric dosage according Bate-Smith method as equivalent of gallic acid. Minerals and salts were quantified by ashes measurement.

Among polyphenols, different flavonol glycosides have been identified by Ultra Performance Chromatography (UPLC) such as quercetin-3-O-rutinoside (rutin), kaempferol-3-O-neohesperoside, kaempferol-3-O-galactoside and kaempferol-3-O-glucoside. Polyphenols namely flavonoids were also detected in different parts of the plant at 405 nm (Fig. 6).

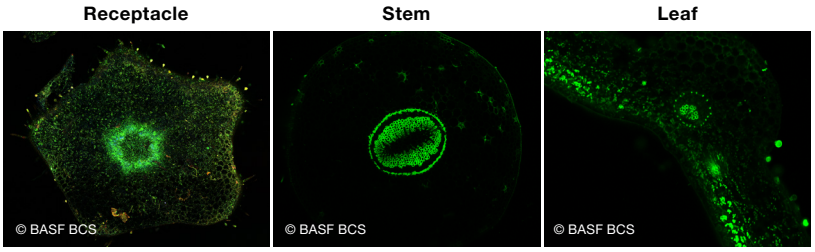


Figure 6 - Polyphenols imaging in aerial part of *L. christinae*.

Safety / tolerability of the product

Oximony was evaluated to ensure its safety in the recommended conditions of use.

Description

Oximony is a preservative free powder, of spray dried *Lysimachia christinae* extract and maltodextrin.

Oximony is an extract of FairWild certified *Lysimachia christinae* aerial parts, characterized in polyphenols, available as a preservative free powder.

INCI name: Maltodextrin (and) Lysimachia Christinae Flower/Leaf/Stem Extract

Recommended dose of use: 0.25%

China IECIC listed



Example: : Illustration of a commercial samples of Oximony with a serum and a mask at 0.25%.



Demonstrated efficacy

The properties of Oximony to promote the skin longevity was studied *in vitro* and *in vivo* on human volunteers.

In vitro the properties of Oximony were demonstrated on:

• Cellular senescence delay in human dermal fibroblasts:
- by decreasing the senescence biomarkers, β-galactosidase activity and p16 level, in a model of replicative senescence,
- by decreasing SASP in a model of stress-induced senescence.

• Pro-taurine action in human epidermal keratinocytes:
- by increasing intracellular taurine content,
- by increasing intracellular taurine transporter TauT content.

• NAD^+ stimulation in human dermal fibroblasts:
- by boosting NAD^+ in a cellular model that mimics skin aging.

• Antioxidant protection and repair:
- by stimulating expression of genes involved in skin protection and repair after oxidative stress, on human dermal fibroblasts and epidermal keratinocytes.

In vivo, we studied in a placebo-controlled clinical study the effects of Oximony at 0.25% on skin fatigue decrease, skin firmness and glow increase on facial skin.

EFFICACY

Skin longevity improvement: inhibition of cellular senescence

in vitro

OBJECTIVE

Accumulation of senescent cells in skin can negatively affect its longevity, functions and regenerative capacities [Franco *et al*, 2022]. Senescent cells present prolonged cell cycle arrest, with increased level of p16 expression, correlated with increased β- Galactosidase activity [Roger *et al*, 2021]. *In vitro*, senescence can be induced by successive multiple cell divisions (replicative senescence) [Hayflick and Moorhead, 1961].

The aim of this study was to assess *in vitro* the effect of Oximony on the replicative senescence in human dermal fibroblast cells.

RESULTS & DISCUSSION

Cells undergoing replicative senescence acquire characteristics of aging cells [Guerrero *et al*, 2021]. Senescence *in vitro* increased over cell culture passage number. Our previous experiments showed that in serial subcultures, the "young" non senescent fibroblasts correspond to 2 until 5 passages and senescent "aged" fibroblasts correspond to more than 10 passages (data not shown). In Oximony evaluation assays, cells were sampled after 5 and 11 passages (P5 and P11) to quantify β-Galactosidase activity and p16 protein level as senescence biomarkers.

At P5, β-Galactosidase activity and p16 expression level quantified by western blot were low. These two senescence biomarkers increased significantly at P11 ($p<0.001$; Fig. 7) showing that the successive passaging of the fibroblast cells in culture have well induced senescence.

TGFβ used as positive control, decreased significantly both β-Galactosidase activity and p16 expression level, thus validating the experiment (data not shown). With Oximony 0.05% at P11, β-Galactosidase activity and p16 protein expression were significantly decreased respectively by 24% (± SEM 7% *vs* untreated control P11 at 100% ± SEM 6%, $p<0.05$) and 34% (± SEM 6% *vs* untreated control P11 at 100% ± SEM 6%, $p<0.01$) compared to the untreated fibroblasts at P11. This evidenced that Oximony delayed replicative senescence of human dermal fibroblast cells.

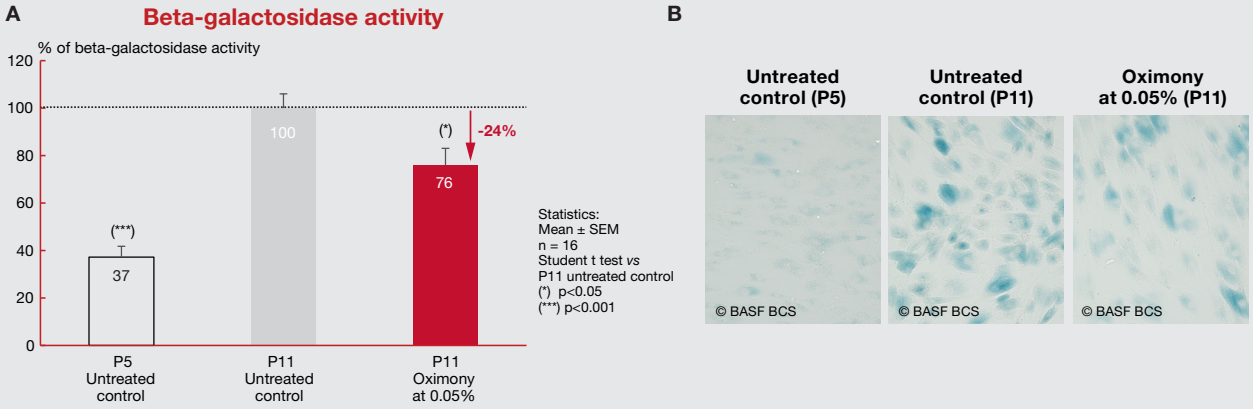


Figure 7 - Quantification (**A**) and visualization (**B**) of enzymatic β-galactosidase activity in senescent fibroblasts.

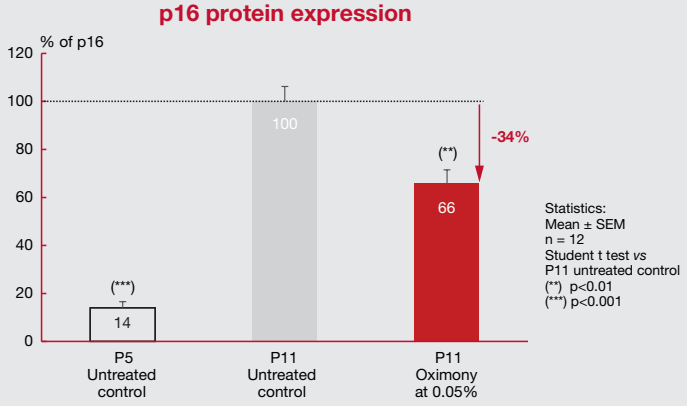


Figure 8 - Quantification of p16 expression in senescent fibroblasts.

CONCLUSION

Oximony decreased significantly the replicative senescence biomarkers β-galactosidase activity and p16 level in human dermal fibroblasts. This contributes to delay skin cell entering in senescence and to increase skin cell longevity.

MATERIALS&METHODS

Study N°EMB160901-2024-051-PROT

Tested products

Culture medium was used as untreated control. TGFβ at 10ng/mL was used as positive control. Freeze dried Lysimachia Christinae Flower/Leaf/Stem Extract (batch PA220124.02) was tested diluted in culture medium at concentration corresponding to Oximony at 0.05%.

Cell culture

Normal human dermal fibroblasts obtained from 18-year-old donor were cultured in monolayers and grown to confluence in defined fibroblast growth medium with antibiotics at 37°C/5% CO_2.

The cells were subcultured every six or seven days. The "young" non-senescent cells are obtained at passage 5 (P5) and senescent cells are obtained with the same donor after culture until passage 11 (P11).

Beta-galactosidase activity quantification

The activity of β-galactosidase in fibroblasts was detected by the senescence Detection Kit (ab65351, Abcam) following the manufacturer instructions. Sixteen replicates were quantified (n=16). Briefly, culture media were removed and cells washed with phosphate buffered saline PBS. The cells were fixed with fixative solution during 15 min at room temperature. Then cells were washed twice with PBS and the staining solution mix was added in each well and incubated overnight at 37°C. The cells were finally observed under microscope (AxioObserver, ZEISS) for development of blue color and color quantification. Cell nuclei were also counted to standardize the β-Galactosidase activity.

p16 quantification by Western blot

The cells were harvested and lysed with a specific lysis buffer. Protein concentration was determined by bicinchoninic assay (BCA protein assay), then all the samples were adjusted to the same protein concentration and deposited at the same quantities in each capillary. Samples and reagents were prepared according to the recommended manufacturer's instructions. Target protein, p16 was identified by a capillary electrophoresis-based protein analysis system (ProteinSimple Sally Sue Instrument, ProteinSimple, San Jose, California, USA) using primary antibodies (anti-p16) and immunoprobed using a horseradish peroxidase-conjugated secondary antibody and chemiluminescent substrate. The resulting chemiluminescent signal was detected and quantified by Compass Software version 5.0.1 (ProteinSimple) in twelve replicates (n=12).

Results and statistics

β-galactosidase activity was standardized to nuclei number of each condition and expressed as mean percentage ± standard error on the mean (n = 16) compared to the untreated control at the same passage, the untreated control at P11 being the highest level of β-galactosidase activity set at 100%.

p16 expression level was expressed as mean percentage ± standard error on the mean (n=12) compared to the untreated control at the same passage, the untreated control at P11 being highest level of P16 level set at 100%.

The statistics were performed according to SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America).

Statistical analysis between conditions has been assessed by a Student t test with threshold of significance set to 5% ($p<0.05$).

EFFICACY

Skin longevity improvement: inhibition of SASP

in vitro

OBJECTIVE

Among many other phenotypic and functional changes, senescence is accompanied by increased secretion of metalloproteinases (MMPs), pro-inflammatory molecules, and other also known as the senescence-associated secretory phenotype (SASP). SASP participates to the induction of senescence into the neighboring cells in a paracrine manner leading to increase in the accumulation of senescent cells [Victorellia *et al*, 2020]. The chemokine signaling CXCR2 receptor is also upregulated in senescent cells and amplify this process of cellular senescence [Roger *et al*, 2021, Zhu *et al*, 2024]. In addition, SASP has negative impact on skin: MMPs degradation of skin extracellular matrix and pro-inflammatory cytokines accelerate the skin aging.

The aim of this study was to evaluate *in vitro* in a model of stress-induced senescence in human dermal fibroblasts the effect of Oximony on SASP, through the measurement of MMP1, MMP3, proinflammatory cytokine IL-6, IL-8 secreted, and of the level of the chemokine receptor CXCR2.

RESULTS & DISCUSSION

Sodium butyrate (NaBu) is a histone deacetylase inhibitor which has been shown to induce senescence-like state of human fibroblasts [Xiao *et al*, 1997]. We used it to induce senescence of the dermal fibroblasts and to stimulate the release of SASP molecules MMP1, MMP3, IL-6, IL-8 and the level of CXCR2.

Inhibition of SASP: MMP1 and MMP3 secretion

On NaBu-induced senescent fibroblasts, Oximony significantly decreased the release of metalloproteinases MMP1 and MMP3. MMP1 release was decreased in a dose response manner with a 68% decrease at 0.025% and a 91% decrease at 0.05% (respectively ± SEM 0% and 1%, p<0.001 *vs* NaBu control normalized at 100%, ± SEM 9%) (Fig. 9A).

MMP3 release was also decreased significantly by 56% and 80% respectively with 0.025% and 0.05% of Oximony (respectively ± SEM 6% and 2%, p<0.001 *vs* NaBu control normalized at 100% ± SEM 4%) (Fig. 9B).

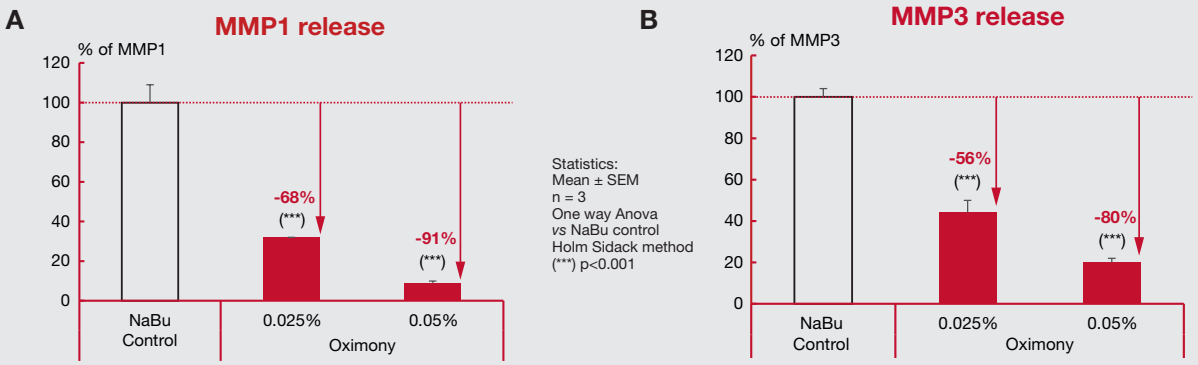


Figure 9 - Quantification of MMP1 (**A**) and MMP3 (**B**) release by fibroblasts in a senescence induced model.

Inhibition of SASP: IL-6 and IL-8 secretion

Oximony also inhibited IL-6 and IL-8 release by fibroblasts in the model of induced senescence (Fig. 10). IL-6 release was inhibited by 82% and 89% with 0.025% and 0.05% of Oximony (respectively ± SEM 1.2% and 2.2%, p<0.001, *vs* NaBu control normalized at 100 ± 2%).

IL-8 release was inhibited by 86% and 95% with 0.025% and 0.05% of Oximony (respectively SEM 0.3% and 0.3%, p<0.001, *vs* NaBu Control normalized at 100%, SEM 1.8%).

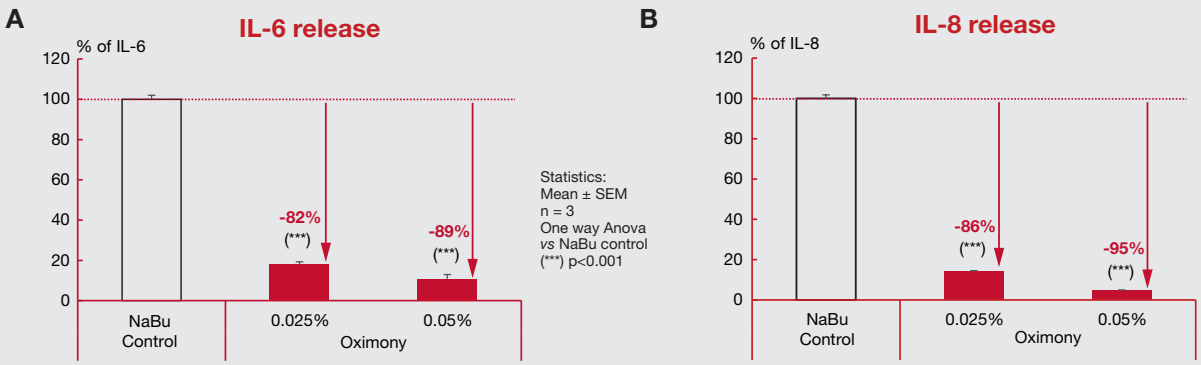


Figure 10 - Quantification of IL-6 (**A**) and IL-8 (**B**) release by fibroblasts in a senescence induced model.

in vivo

Inhibition of CXCR2 production

The production of the chemokine receptor CXCR2 was significantly reduced in a senescence induced model by 60% and 66% with Oximony at 0.025% and 0.05% (respectively SEM 1.0% and 5.8%, p<0.001, *vs* NaBu control normalized at 100, ± SEM 1.5%,) (Fig. 11).

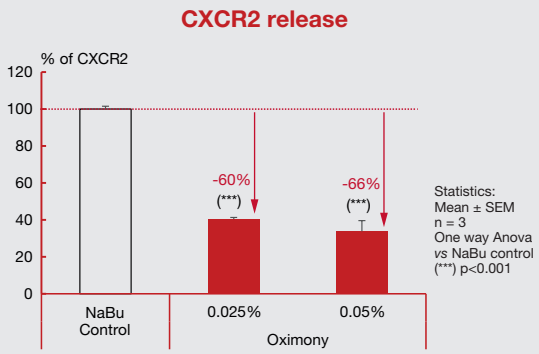


Figure 11 - Quantification of CXCR2 production by fibroblasts in a senescence-induced model.

CONCLUSION

Oximony showed a senostatic effect by decreasing significantly the release of the SASP metalloproteinases MMP1, MMP3 and of cytokines IL-6, IL-8, as well as the level of the CXCR2 receptor overexpressed in senescent fibroblasts. This contributes to decrease the spread of senescence to the surrounding cells and to protect skin from negative effects of SASP and thus to maintain skin longevity.

MATERIALS&METHODS

Study N°EMB160901-2024-034-CC

Tested products

Freeze-dried Lysimachia Christinae Flower/Leaf/Stem Extract (batch PA220124.02) was tested diluted in culture medium at concentrations corresponding to Oximony at 0.025% and 0.05%.

Cell culture

Normal human dermal fibroblasts obtained from 34-year-old donor were cultured in monolayers and grown to confluence in defined fibroblast growth medium with antibiotics at 37°C/5% CO_2.

Treatment

Sodium Butyrate (NaBu) was used as senescence inducer. Cells were treated 24h with the tested compound, then treated 48h with NaBu 20mM and the tested compound. Culture medium with NaBu was used as control (NaBu control).

MMP1 and MMP3 quantification

MMP1 and MMP3 were quantified on cell culture supernatants with MMP1 and MMP3 specific ELISA kit from Millipore (Saint Louis USA). Briefly, 100 µl of each standard and sample were deposited into appropriate wells and incubated 2.5 hours at room temperature or overnight at 4°C with gentle shaking. Then solution was discarded, and the wells washed and incubated 1 hour with detection antibody at room temperature, followed by 100 µl of prepared Streptavidin solution to each well; After 45 minutes of incubation, the solution was discarded and the wells washed. Then 100 µl of TMB One-Step Substrate Reagent were added to each well. After 30 minutes of incubation at room temperature, 50 µl of Stop Solution were added to each well. The absorbance was recorded at 450 nm immediately and MMP1 and MMP3 quantities were deducted from the corresponding standard curve.

IL-6 and IL-8 quantification

IL-6 and IL-8 were quantified on cell culture supernatant with specific ELISA kit from R&D Systems (USA).

This assay employs the quantitative sandwich enzyme immunoassay technique. Monoclonal antibody specific for human IL-6 and IL-8 have been pre-coated onto a microplate. Standards and samples are pipetted into the wells and any IL-6 present is bound by the immobilized antibody. After washing away any unbound substances, an enzyme-linked polyclonal antibody specific for human IL-6 or IL-8 are added to the wells. Following a wash to remove any unbound antibody-enzyme reagent, a substrate solution is added to the wells and color develops in proportion to the amount of IL-6 bound in the initial step. The color development is stopped, and the intensity of the color is measured.

CXCR2 quantification

CXCR2 was quantified on cell culture supernatant. The ELISA kit from Novus biological uses the Sandwich-ELISA principle. The micro-ELISA plate has been pre-coated with an antibody specific to Human CXCR2. Standards or samples are added to the micro-ELISA plate wells and combined with the specific antibody. Then a biotinylated detection antibody specific for Human CXCR2 and Avidin-Horseradish Peroxidase (HRP) conjugate are added successively to each micro plate well and incubated. Free components are washed away. The substrate solution is added to each well. Only those wells that contain Human CXCR2, biotinylated detection antibody and Avidin-HRP conjugate will appear blue in color. The enzyme-substrate reaction is terminated by the addition of stop solution and the color turns yellow. The optical density (OD) is measured spectrophotometrically at a wavelength of 450 nm. The OD value is proportional to the concentration of Human CXCR2. Human CXCR2 quantity in the samples was obtained by comparing the OD of the samples to the standard curve.

Results and statistics

Results are expressed as Mean value ± SEM (n=3) compared to the NaBu control set as 100%

The statistics were performed according to SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America).

Statistical analysis between NaBu control and Oximony was assessed by One Way Anova test with threshold of significance set to 5% (p<0.05).

EFFICACY

Skin longevity improvement: increase in cellular taurine and taurine transporter TauT content

in vitro

OBJECTIVE

Taurine molecule is able to target and inhibit several hallmarks of aging including cellular senescence. In mammal, supplementation with taurine showed promising effect on health and longevity [Singh *et al*, 2023]. Furthermore, taurine and its transporter have been shown to be present in epidermis [Siefken *et al*, 2003].
The aim of the study was to evaluate *in vitro* the stimulatory effect of Oximony on intracellular taurine and transporter Taut content on human epidermal keratinocytes.

RESULTS & DISCUSSION

Effect of Oximony on taurine and TauT content in keratinocytes were quantified using ELISA method. In this model, phorbol myristate acetate at 10 ng/mL was used as a positive control of experimentation as it has been previously shown to increase cellular content in taurine and its transporter [Kim HW *et al*, 2003]. In our study, it increased significantly taurine and TauT content in keratinocytes thus validating the experiment (data not shown).

In our experimental conditions, Oximony at 0.025% and 0.05% increased significantly taurine content in keratinocytes by 44% and 38% compared to the untreated control (respectively ± SD 5.5% and 12.3%, p<0.01 *vs* the Untreated control normalized at 100 ± SD 19.8%) (Fig. 12).

Regarding TauT level, Oximony at 0.025% and 0.05% increased it significantly and in a dose response manner by 117% and 162% compared to the untreated control (respectively ± SD 9.4% and 53%, p<0.001, *vs* the untreated control normalized at 100 ± SD 8.0%) (Fig. 13).

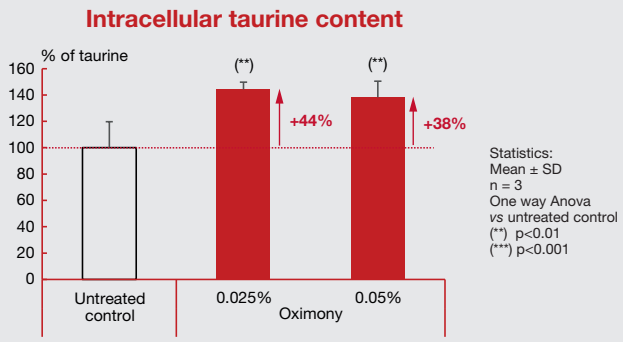


Figure 12 - Intracellular taurine quantification in keratinocytes.

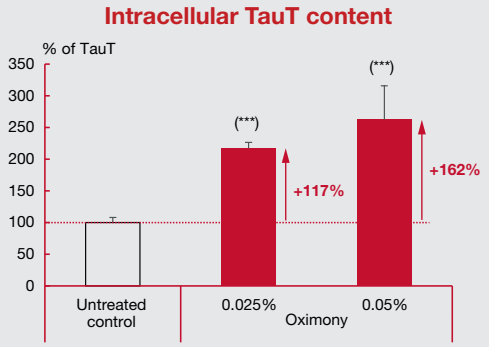


Figure 13 - Intracellular TauT quantification in keratinocytes.

CONCLUSION

Oximony increased significantly the cellular level of taurine and its transporter TauT in human epidermal keratinocytes and may thus contribute to enhance skin longevity.

MATERIALS&METHODS

Study N°EMB160901-2022-764-PROT

Tested products

Freeze-dried Lysimachia Christinae Flower/Leaf/Stem Extract (batch PA220124.02) was tested at concentrations corresponding to Oximony at 0.025% and 0.05%.

Cell culture

Human epidermal keratinocyte cells were obtained from a 39-year-old donor. They were cultivated in keratinocyte medium until confluency at 37°C and 5% of CO_2.

Treatment

Cells were incubated during 24 hours in absence or in presence of the tested compound diluted directly in incubation medium. Culture medium was used as untreated control. Phorbol myristate acetate at 10ng/mL was used as positive control.

Taurine quantification

At the end of the incubation period, intracellular taurine content was quantified in cell lysate, obtained by sonication, using a specific ELISA kit.

TauT quantification

At the end of the incubation period, intracellular TAUT content was quantified in cell lysate, obtained by sonication, using a specific ELISA kit.

Protein quantification

At the end of the incubation period, the protein level in the cell lysates was quantified using a Spectrocolorimetric method (Bradford method).

Results and statistics

The results are expressed in percentage of taurine and TauT normalized at the level of cellular proteins compared to the untreated control set at 100%. The statistics were performed according to SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America). Statistical analysis between “Control” and Oximony has been assessed by a One Way Anova test followed by a Holm-Sidak test, with threshold of significance set to 5% (p<0.05).

EFFICACY

Skin longevity improvement by boosting NAD⁺, a skin longevity factor

in vitro

OBJECTIVE

Nicotinamide adenine dinucleotide (NAD^+) is an essential co-factor that serves to mediate various biological processes, including metabolism, DNA repair, and gene expression [Canto *et al,* 2015; Magni *et al,* 2004]. Recent studies have highlighted the various roles of NAD metabolism in aging and longevity. It is indicated that NAD levels decrease with age due to an alteration in the balance between NAD synthesis and consumption. Decreased NAD levels are associated with several aging-related disease, including metabolic diseases, cancer, and neurodegenerative diseases. Accordingly, the dietary administration of NAD precursors has been shown to re plenish NAD levels in aged tissues and exhibit beneficial effects against aging and aging-related diseases [Fang *et al,* 2017; Rajman *et al,* 2018; Yoshino *et al,* 2017]. Most importantly, boosting NAD metabolism has been shown to extend the lifespan of various organisms, such as yeast, worms, flies, and rodents.

As a co-enzyme, NAD is involved in numerous enzymatic reactions. NAD plays a central role in the regulation of energy metabolism pathways, including glycolysis, fatty acid oxidation (β oxidation), the tricarboxylic acid (TCA) cycle, and oxidative phosphorylation [Canto *et al,* 2015]. It is a crucial cofactor for sirtuins, a family of proteins that function as NAD^+-dependent deacetylases. Sirtuins are involved in regulating metabolism, stress responses, aging, and longevity [Imai *et al,* 2016].

The aim of this study was to assess *in vitro* the effect of Oximony on NAD^+ stimulation on normal human fibroblasts.

RESULTS & DISCUSSION

The NAD^+ production and maintenance critically require the activity of Nicotinamide phosphoribosyltransferase (NAMPT) involved in “NAD salvage pathway”, the major pathway for maintaining cellular NAD levels. In this pathway, the synthesis of NAD starts from nicotinamide NAM which is a precursor already present in the cell; it is a by-product of NAD-consuming enzymes such as sirtuin and is also imported as a dietary nutrient from various foods. NAM is then converted to NAD through a two-step reaction; first by the enzyme NAMPT, which generates nicotinamide mononucleotide NMN which is secondly converted in NAD by NMNAT enzyme [K Yaku *et al,* 2018].

The level of NAMPT is described to decline in aged tissues of rats and mice. In human cells, NAMPT overexpression increases intracellular NAD^+ levels and has been shown to extend replicative lifespan in human aortic endothelial cells and human smooth muscle cells [Khaidizar *et al,* 2017]. The use of FK866 (an inhibitor of NAMPT) to investigate the effect of the NAD salvage pathway inhibition is also described in the literature [Gerner *et al,* 2018]. To mimic the age-related decrease in NAMPT levels, we use FK866 to inhibit NAMPT.

The positive control NMN, tested at 2mM induced a strong stimulation of NAD^+ (data not shown). These results were expected and validated the assay.

The stress molecule FK866 tested alone at 1µM significantly inhibited NAD^+ by 232% (SEM ± 26%, $p<0.001$ vs untreated control). In presence of Oximony at 0.05%, NAD^+ was significantly stimulated by 74% (SEM ± 31%, $p<0.01$ vs stressed control). This evidenced that Oximony contributes to boost NAD^+ in a cellular model that mimics skin aging.

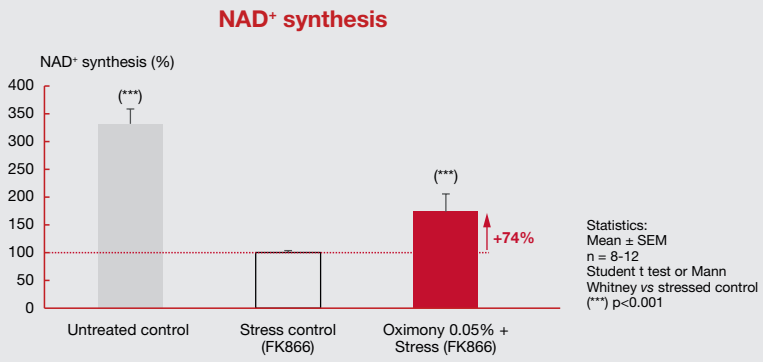


Figure 14 - Effect of Oximony on NAD^+ synthesis.

CONCLUSION

Oximony significantly boosts NAD^+ in normal human fibroblasts. By stimulating this skin longevity factor, Oximony contributes to skin longevity improvement.

MATERIALS&METHODS

Study N° TME-2023-010-2024_NHDF_FK866_NAD^+

Tested products

Culture medium was used as untreated control.

FK866 hydrochloride hydrate (F8557-5MG, Sigma) at 1µM was used as a stress molecule.

Nicotinamide mononucléotide NMN (Sigma N3501) at 2mM was used as positive control of NAD^+ stimulation.

Oximony BC10187 (batch 0029121889) was tested at 0.05% diluted in culture medium.

Cell culture

Normal human dermal fibroblasts (NHDF), obtained from healthy donors, were seeded in 96-well plates at a rate of 50.000 cells/cm^2 and cultured in DMEM/Ham's F-12 medium (5:1), supplemented with 10% FCS and 0.5% antibiotic for 24h at 37°C, under 5% CO2. Then cells were incubated during 24h with the product or with positive control or without any product (untreated control) in DMEM supplemented with 0.5% antibiotic and 1µM of FK866 hydrochloride hydrate.

The lack of cytotoxicity of the treatments was checked thank to the Thiazolyl Blue Tetrazolium Bromide (MTT) reagent. Briefly, after washing, cells were incubated with MTT at 0.25 mg/mL for 2h30 at 37°C. Then, the cells were washed and incubated with 1mL of Dimethylsulfoxide (DMSO). The absorbance was read at 540nm.

NAD^+ assay by an enzymatic test

Finally, the NAD^+ release in supernatant medium was measured by a specific commercial enzymatic method (MAK460-1KT, Sigma) according to the supplier's instructions. The principle is based on a lactate dehydrogenase cycling reaction in which the formed NAD^+/NADH reduces a probe into a highly fluorescent product. The fluorescence intensity of this product, measured at λEx = 530 nm/λEm = 585 nm, is proportional to the NAD^+/NADH concentration in the sample.

Results and statistics

The results are expressed as the mean percentage of NAD^+ level (%) ± standard error of mean (SEM) (n=8-12) and compared to the FK866 stressed condition standardized at 100%.

All statistics were performed according to SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America). The statistical evaluations were calculated using Student t test or Mann-Whitney test vs FK866 stressed condition with NS: $p>0.05$, (*) $p<0.05$, (**) $p<0.01$, (***) $p<0.001$.

EFFICACY

Anti-oxidant defense and repair

in vitro

OBJECTIVE

Oxidative stress leads to biological macromolecules (protein, lipids, DNA) damages and accelerates cellular senescence and aging process.
The aim of this study was to evaluate the effect of Oximony on cellular antioxidant pathways through PCR gene array followed by quantitative PCR on human epidermal keratinocytes and dermal fibroblasts.

RESULTS & DISCUSSION

Gene expression analysis by PCR arrays is considered to be one of the most effective methods for studying a panel of targeted genes. A PCR array focused on 91 genes involved in the cell defense against oxidative stress was specifically designed. This method enabled us to draft an initial "profile" of activity and to identify the main modulated antioxidant pathways. Resveratrol was used as positive control to validate the assay with the upregulation of HMOX1 (for fibroblasts) and SOD1 (for keratinocytes).

The PCR array showed that Oximony modulated the expression of many genes involved in skin defense against oxidative stress (Fig. 15). PCR array showed up-regulations of more than ten genes in fibroblasts and in keratinocytes.

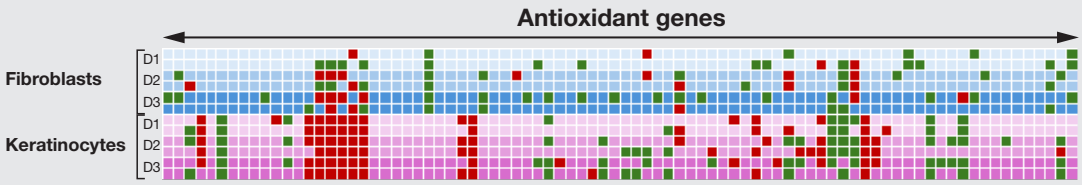


Figure 15 - Results of the PCR array on antioxidant defense and DNA repair gene expression modulated by Oximony in human dermal fibroblasts and epidermal keratinocytes.
Oximony tested at 3 doses (D1= 0.0125%; D2=0.025%; D3=0.05%). Red: Downregulation <-1.5. Green: Overexpression > 1.5 / control.

Then, for each of the identified upregulated genes in the PCR array, expression levels were confirmed by qRT-PCR independently on each of 7 fibroblasts donors and 4 keratinocytes donors cultures. The average upregulation fold changes are presented in Table 1 and 2.

Gene symbol	Gene function	Upregulation
SESN2	Regulates positively the transcription of genes involved in the response to oxidative stress. Attenuates cellular senescence	x 1.3 (*)
PRDX3	Plays a role in cell protection against oxidative stress by detoxifying peroxides	x 1.3 (*)
SOD2	Encodes a mitochondrial protein that binds to the superoxide bypro-ducts of oxidative phosphorylation and converts them to hydrogen peroxide and diatomic oxygen	x 4 (*)
GPX3	Encodes a protein of glutathione peroxidase family, which catalyze the reduction of organic hydroperoxides and hydrogen peroxide (H_2O_2) by glutathione, and thereby protect cells against oxidative damage.	x 2.2 (*)
OXR1	Mitochondrion and Nucleolus genes that encodes for protein involved in resistance to oxidative stress	x 1.5 (*)
GADD45B	Transcript levels are increased following stressful growth arrest condi-tions and treatment with DNA-damaging agents. Encodes for a protein which responds to environmental stresses by mediating activation of the p38/JNK pathway via MTK1/MEKK4 kinase	x 1.6 (*)
AKR1B1	Catalyzes the NADPH-dependent reduction of a wide variety of carbo-nyl-containing compounds to their corresponding alcohols	x 1.2 (*)

Table 1 - Upregulation of antioxidant genes by Oximony at 0.05% in human dermal fibroblasts by qRT-PCR. Mean value, n=7, Student t test or Mann-Whitney test. (*) p<0.05.

in vitro

In fibroblasts, genes were upregulated after 48h of treatment except the PRDX3 which was upregulated after 24h of treatment.

Among the upregulated genes some of them are directly involved in senescence modulation.

The proteins coded by some upregulated genes have been already evidenced to attenuate development of cellular senescence; expression of GADD45B, a protein involved in DNA damage repair and cell cycle control, can attenuate the development of senescence [Zaidi *et al*, 2022].

Induction of peroxiredoxins such as peroxiredoxin 3 was also known to partially reverse senescence [Jang *et al*, 2020].

The Sestrin 2 protein, whose expression is associated with DNA damage, is also known to attenuate cell senescence. These proteins protect the regenerative potential of stem cells against stress, modulate their cell viability and ability to differentiate [Haidurov *et al*, 2020]. At cellular level, Sestrins suppress cell senescence in fibroblasts, protecting them from aging by suppressing SASP in various tissues [Haidurov *et al*, 2020]. Sestrin 2 is a regulator of cell senescence, its depletion leading to cell entry into senescence [Hwang *et al*, 2020].

Gene Symbol	Gene function	Upregulation
HMOX1	Protects against programmed cell death based on its ability to catabolize free heme	x 3.7 (*)
SQSTM1	Mediates activation of NF-kB in response to upstream signals	x 3 (*)
AKR1C1	Catalyzes the conversion of aldehydes and ketones to their corresponding alcohols by utilizing NADH and/or NADPH as cofactors	x 3.6 (*)
GSTM3	Functions in the detoxification of electrophilic compounds, including carcinogens, environmental toxins and products of oxidative stress, by conjugation with glutathione	x 1.7 (*)
PRDX5	Protects cells against oxidative stress by detoxifying peroxides	x 1.8 (*)
SEPP1	Encodes an extracellular antioxidant. Involved in selenium transport	x 2.1 (*)
GADD45A	Transcript levels are increased following stressful growth arrest conditions and treatment with DNA-damaging agents. The protein encoded by this gene responds to environmental stresses by mediating activation of the p38/JNK pathway via MTK1/MEKK4 kinase	x 1.5 (*)
GADD45B		x 1.5 (*)
GADD45G		x 2.1 (*)

Table 2 - Upregulation of antioxidant genes by Oximony at 0.05% in human epidermal keratinocytes by qRT-PCR. Mean value, n=4, Student t test or Mann-Whitney test. (*) $p<0.05$.

In keratinocytes, genes were upregulated after 24h of treatment except the GSTM3 which was upregulated after 48h of treatment.

CONCLUSION

Oximony upregulated the expression of 7 antioxidant and DNA repair genes in human dermal fibroblasts and of 9 antioxidant DNA repair genes in human epidermal keratinocytes, helping thus to enhance the antioxidant cell defense in the skin.

MATERIALS&METHODS

Study N°EMB160901-2023-796-BM

Tested products

Freeze-dried Lysimachia Christinae Flower/Leaf/Stem Extract (batch PA220124.02) was tested diluted in culture medium at concentrations corresponding to Oximony 0.0125%, 0.025% and 0.05%.

Cell culture

4 strains of human fibroblasts for PCR array and 7 strains for qPCR validation were cultured in complete HNDF medium for 3 days.

3 strains of human keratinocytes for PCR array and 4 strains for qPCR validation were cultured in MCEC medium for 1 day then 3 days in MCDB153 culture medium. All cultures were made at 37°C and 5% of CO_2.

Treatment

Cells were incubated during 24 and 48h hours in absence (untreated control) or in presence of the tested compound. Resveratrol at 50µM (solubilized in ethanol) was used as positive control (data not shown) to validate the assay.

PCR array

Total RNAs from 4 cultures were extracted with NucleoSpin kit (Machery-Nagel), according to the manual instructions. Quantification and quality control of RNAs were performed by measurement of optical densities at 260 and 280 nm. Reverse transcriptions were realized on 0.1 µg of RNA (extracted from the 4 cultures) in 20 µl final mix using SuperScript IV VILO Master Mix Kit (Invitrogen) and the C100 Touch Thermocycleur (BioRad) for each condition. Quantitative PCRs were performed on 2 µl of RT mix, diluted at 1/2 in 10 µl final reaction, according to manual instruction (LC 480 SYBR Green I Master, Roche) in a LC480 thermocycler (Roche). Each analysis was performed in duplicate, and the mean used to calculate gene expression compared to a housekeeping gene (SDHA: Succinate Dehydrogenase Complex Flavoprotein Subunit A) analyzed in duplicate. Reverse transcriptions were performed on 1.5 µg of RNA from 4 cultures (1/1) in 20 µl final mix, using iScript Advanced cDNA Synthesis Kit (BioRad) in the C100 Touch Thermocycleur (BioRad). PCR-array were performed on 1 µl of RT mix, diluted at 1/10 in 10 µl final reaction, according to the manuel instruction (SsoAdvanced Universal SYBR Green Supermix, BioRad) in a PrimePCR Custom Plate 384 Well (SLO-ROS).

Each analysis was performed in duplicate and the mean used to calculate gene expression compared to the mean of 3 housekeeping genes (ACTB: Actin B; PPIA: Peptidylprolyl Isomerase A; ALAS1: 5'-Aminolevulinate Synthase 1) analyzed in duplicate. Obtained results for Oximony-treated conditions were compared to the untreated condition and the calculated mean conserved as PCR-array results.

qPCR validation

Reverse transcriptions were realized on 0.1 µg of RNA (extracted from each of the 7 cultures for fibroblasts and 4 cultures for keratinocytes) in 20 µl final mix using SuperScript IV VILO Master Mix Kit (Invitrogen) and the C100 Touch Thermal Cycler (Bio-Rad) for each condition. Quantitative PCRs were performed on 2 µl of RT mix, diluted at 1/2 in 10 µl final reaction, according to manual instruction (LC 480 SYBR Green I Master, Roche) in a LC480 thermocycler (Roche). Each analysis was performed in duplicate and the mean used to calculate gene expression compared to the mean of 2 housekeeping genes (ACTB: Actin B; PPIA: Peptidylprolyl Isomerase A) analyzed in duplicate. Results of the 7 cultures for fibroblasts and 4 cultures of keratinocytes were used for statistical analysis.

Results and statistics

Results were calculated as a fold change of gene expression in the Oximony treated condition compared to the untreated condition for each culture (n=4 or n=7) and expressed as the mean (± standard error of the mean, data not shown). The statistical analysis was carried out via the Student t test or Mann et Whitney *vs* the untreated condition. The threshold of significance was set to 5% ($p<0.05$).

EFFICACY

Improvement of skin aging and longevity: skin fatigue, firmness and glow

in vivo

OBJECTIVE

With the progression of the age, dull skin appears [Eggerstedt *et al*, 2023], skin firmness decreases while skin fatigue increases, [Henning *et al*, 2022; Dobrev *et al*, 2005].
The aim of the clinical study was to evaluate the efficacy of Oximony at 0.25% as compared to a placebo on improvement of these signs of skin aging (Fig. 16).
The study was a double-blind, placebo controlled, whole face and randomized study of 62 Chinese female subjects divided in two groups, aged from 40 to 55 years old.

Randomized whole face, twice daily self-applications for 56 days.

Placebo
(31 volunteers - Mean age 47 yo)



Oximony 0.25%
(31 volunteers - Mean age 48 yo)



The efficacy of Oximony at 0.25% was evaluated.
Evaluation of skin fatigue (Cutometer) on the cheek, of skin firmness (Indentometer) and skin glow (Glossymeter) on the front face. Pictures were taken on the face and a self assessment was done
Time point measurement: D0, D14, D28 and D56.

Figure 16 - Clinical study design.

RESULTS & DISCUSSION

Effect on skin fatigue

After 14 days of application, Oximony at 0.25% showed a significant increase ($p<0.001$) of 28% in F3 as compared to baseline, up to 44% ($p<0.001$) after 28 days and 37% ($p<0.01$) after 56 days of treatment. These increases were significant *vs* the placebo at all the time points, by 36% ($p<0.001$) after 14 days, by 30% ($p<0.001$) and 26% ($p<0.01$) after 28 and 56 days of application (Fig. 17 and 18).

These results evidence the capacity of Oximony at 0.25% to strongly decrease skin fatigue starting after 14 days and continuing after 28 and 56 days of application.

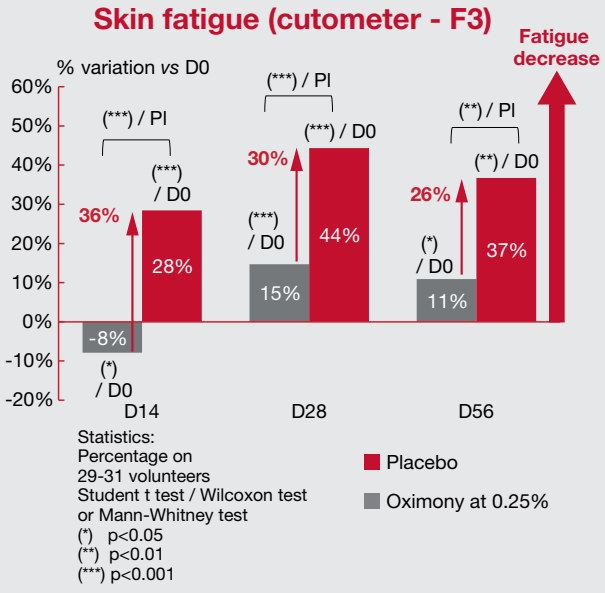


Figure 17 - Measurement of skin fatigue by cutometer (F3 Parameter) after Oximony application.

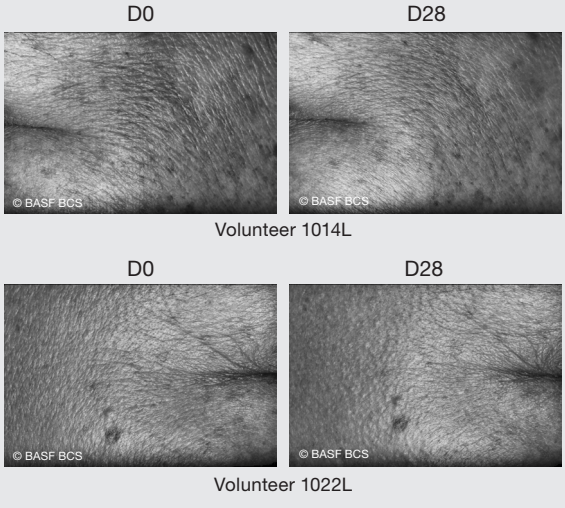


Figure 18 - Illustrative images after Oximony application.

in vivo

Effect on skin firmness

Starting at 28 days of application, Oximony at 0.25% decreased significantly depth into the skin surface compared to the placebo (-6.1%, p<0.01), and by 17% compared to the baseline (p<0.001). This significant difference was maintained after 56 days of application with a significant decrease by 7.3% compared to placebo (p<0.05) and by 12% (p<0.001) compared to baseline (Fig. 19).
After 14 days, there was already a decrease *vs* the baseline, by 8.8% (p<0.001), but the difference *vs* placebo did not reach statistical significance.
These results demonstrate the ability of Oximony at 0.25% to increase the firmness of the skin, with first benefits measurable after only 14 days.

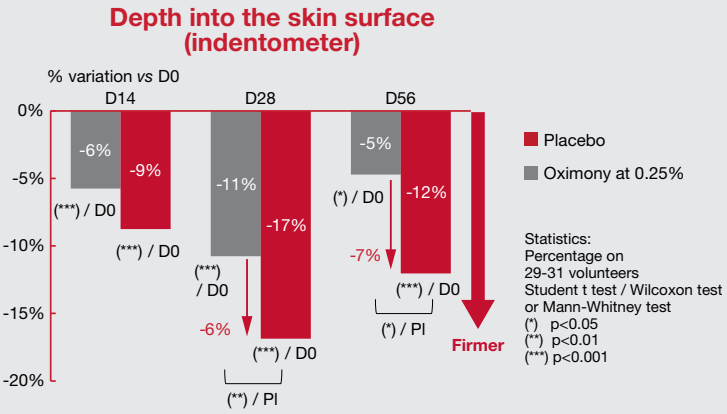


Figure 19 - Measurement of skin firmness by indentometer after Oximony application.

Effect on skin glow

After 56 days of application Oximony at 0.25% increased significantly the skin glow by 6.2% (p<0.001) compared to the placebo and by 18% compared to baseline (p<0.001). After 14 days and 28 days, glow was already increased *vs* the baseline, respectively by 7.4% (p<0.05) and 9.6% (p<0.01), but the difference *vs* placebo did not reach statistical significance.
These results show capacity of Oximony at 0.25% to increase skin glow (Fig. 20 and 21), with first benefits measurable after only 14 days.

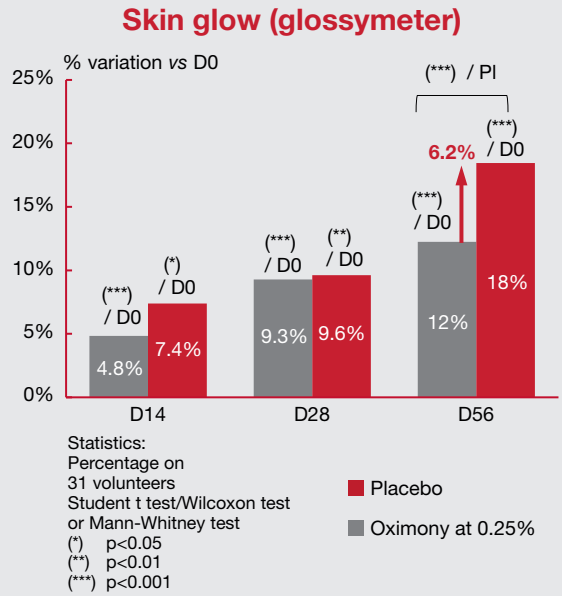


Figure 20 - Measurement of skin glow by glossymeter after Oximony application.

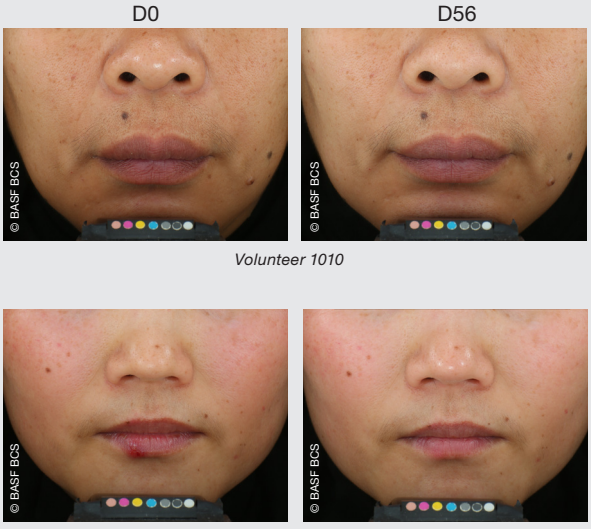


Figure 21 - Illustrative pictures of skin glow after Oximony at 0.25% application.

Self-assessment by volunteers

Volunteers' self-assessment confirmed the instrumental measurement. After the first 14 days of application of the 0.25% Oximony gel cream, a statistically significant majority (p<0.001) perceived that their skin felt smoother (94%), more refined (94%), firmer (84%), soft and supple (90%), looked glow (94%) and more even (90%).

These results are confirmed at the other timepoint.

CONCLUSION

With measurable and perceived effects as early as 2 weeks of application, Oximony demonstrated an improvement of skin fatigue, firmness and glow.

MATERIALS&METHODS

Study N°EMB160901-10162

Tested products

Lysimachia christinae BC10187 batch PA221024.01.

Study design

The clinical study was carried-out randomized, double-blind, and on whole face vs placebo. The efficacy of the gel cream emulsion containing Oximony at 0.25% was compared to the baseline (before treatment, D0) and to the placebo gel cream emulsion (same formulation without Oximony). The gel cream formulations are detailed in Annex 2. The study was conducted during a period of 56 days with check points at D0, D14, D28 and D56.

Application modality

Volunteers applied twice a day, once in the morning and once in the evening, for 56 days the Oximony emulsion at 0.25% or the placebo emulsion on whole face, under normal conditions of use. The last application was done the evening before the visit.

Inclusion criteria

62 Chinese female volunteers, aged from 40 to 55 years old, with crow's feet wrinkles. Volunteers were divided in two groups; one applied the placebo (31 volunteers; mean age 47 yo) and the other applied Oximony at 0.25% (31 volunteers; mean age 48 yo).

Evaluation methods

Skin fatigue measurement by Cutometer

The Cutometer dual MPA 580 was used to measure the skin fatigue. The measuring principal is based on the suction method. Negative pressure is created in the device, and the skin is drawn into the aperture of the probe and released again after a defined time. Inside the probe, the penetration depth is determined by a non-contact optical measuring system. This optical measuring system consists of a light source and a light receptor, as well as two prisms facing each other. The light intensity varies according to the penetration depth of the skin.

With Cutometer, the skin fatigue F3 parameter corresponds to the measure of the skin's capacity for successive deformation and relaxation. The higher the value, the lower the skin fatigue.

The measurements were done on both sides of the cheekbone skin

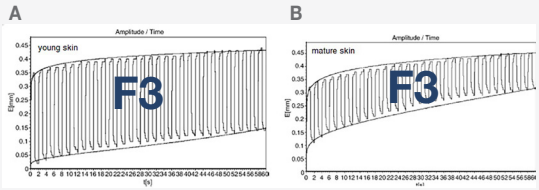


Figure 22 - Representation of skin fatigue parameter F3: deformation and return amplitude over time. (A) Young and tonic skin - (B) Aged and tired skin.

When the skin fatigue increases the skin progressively loses its capability to restore its initial position and each subsequent curve has lower amplitude and lower elastic retraction (Fig. 20). This results in a decrease in F3 which is the area within the envelope curves (maximum and minimum) of successive deformation and relaxation [Dobrev H, 2005].

Skin firmness measurement by Indentometer

The measurement principle is based on the force (by a spring) used on the small indenter of the probe to deform the skin. The device measures the penetration depth of the probe indenter (pin) into the skin in mm. The more the small indenter pin can deform the skin with a given pressure, the less firm the skin is.

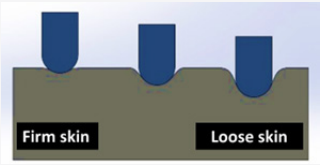


Figure 23 - Indentometer measurement principle. The measurements were done on the front face.

Skin glow measurement by Glossymeter

The Skin-Glossymeter GL 200 xas used to measure the skin gloss. The measurement of gloss is based on reflection of light sent to the skin. Parallel white light is created by LEDs in the probe head. To be able to emit light at 60° in a relatively small and uniquely designed measurement head, light is sent out at 0° and reflected by a mirror to 60°. Two separate measurement channels measure the reflected light.

Direct reflection/gloss: the direct reflected light is again guided by a mirror in 60° into the reflection channel (blue).

Diffuse reflected/scattered light: The scattered/diffuse reflected light is measured at 0° (completely vertically above the measured surface) under the assumption that light is scattered in the same way over all degrees (diffuse channel - red). So, the Skin-Glossymeter GL 200 expresses both, the portion of directly reflected light (gloss) and the diffuse scattered portion from the skin surface.

The results are expressed in (arbitrary) Glossymeter units. The greater the value, the more the skin glows.

The measurements were done on both sides of the front face.

Illustrative picture

Illustrative photographs were taken before the study D0, at D14, D28 and at D56 the end of the treatment in color and black and white. The photographs were done on whole face (Front face, 45° to the left and 45° to the right).

The photographs have been cropped to highlight areas of interest: Crow's feet and temple for fatigue and lower face for the glow

Self-assessment

The subjects were asked to answer self-evaluation questionnaire at D14, D28 and D56 visit, to evaluate the overall opinion and their attitude towards the efficacy of the products under test.

For each item, the possible answers are 1 "Strongly disagree", 2 "Disagree", 3 "Neither agree nor disagree", 4 "Agree" and 5 "Strongly agree".

The evaluation was performed on whole face.

For each item, the % of favorable answer (4 "Agree" + 5 "Strongly agree") were calculated.

Results and statistics

Results are expressed as the mean percentage of change compared to the baseline measurement (D0).

Instrumental values

The statistical analysis of the parameters has been done after the verification of the normality of distribution using Shapiro-Wilk test.

The comparison between the different times and the baseline time (D0) being made on the same volunteer (paired comparison), the following tests were used:

- validation of the normality of the studied parameters: Student t test,
- invalidation of the normality of the studied parameters: Wilcoxon test.

The treatments were applied by 2 different groups of volunteers (unpaired comparison), the following tests were used:

- validation of the normality of the studied parameters: Student t test,
- invalidation of the normality of the studied parameters: Mann-Whitney test.

Consumer self-perception questionnaire

The analysis involves establishing frequency tables that take into account the number of responses and calculating the frequency of the different possible answers (given as percentage) to each qualitative question.

To evaluate the efficacy and the appreciation of the products for each item, two percentages Z1 and Z2 are calculated as follows:

Z1 = favorable opinion (4 "Agree" and 5 "Strongly agree").

Z2 = unfavorable opinion (1 "Strongly disagree", 2 "Disagree ").

The statistical difference in frequencies (%) between favorable (agreement) and unfavorable (disagreement) opinions is evaluated using Two-tailed binomial test at 5%.



GENERAL CONCLUSION

Longevity Hero for skin that defies time

- Pro-Taurine longevity cutting-edge ingredient that embraces the longevity trend.
- Natural extract of sustainably sourced and FairWild certified *Lysimachia christinae*.
- Stimulates *in vitro* taurine natural production by epidermal cells, delays skin cell senescence and boosts NAD^+ synthesis.
- Clinically proven to decrease skin fatigue and increase skin firmness and glow.

Beauty now focuses on extending the lifespan of the skin, not just by combating visible signs of aging, but by enhancing longevity at every level. Our relationship with aging is evolving, reflected in the shift of cosmetic products towards longevity beauty, replacing the old anti-aging narrative of the 2000s.

As part of our Responsibly Active program, Oximony embodies a commitment to sustainability and ethical sourcing. It is a natural extract of sustainably sourced and FairWild certified *Lysimachia christinae* (INCI: Maltodextrin (and) Lysimachia Christinae Flower/Leaf/Stem Extract).

In vitro evaluation of performance evidenced that Oximony was able to delay cellular senescence in a model of replicative senescence. Additionally, Oximony reduces Senescence-Associated-Secretory-Phenotype (SASP) secretion with decrease of MMP1, MMP3, IL-6 and IL-8 and of the senescence-amplifying chemokine receptor CXCR2 in a senescence-induced model.

Moreover, Oximony showed *in vitro* a pro-taurine action, by increasing intracellular taurine content in epidermal keratinocytes as well as its transporter TauT necessary for its diffusion in and out cells. Complementarily, Oximony supports cellular anti-oxidative defense. It increased *in vitro* the expression of several genes involved in the protection against oxidative stress and DNA damages and may thus limit oxidative stress induced senescence.

Tested *in vivo* for 56 days in a placebo-controlled clinical study on 62 volunteers, Oximony at 0.25% decreased significantly skin fatigue by 36% compared to the placebo after just two weeks, and reached 44% of improvement compared to baseline after one month of application. Additionally, compared to baseline with significant difference *vs* the placebo, it boosted skin firmness by 17% after one month and increased glow by 18% after two months, with noticeable improvements observed as early as two weeks after application. These performances on skin aspect, firmness and glow after just two weeks of use were perceived by 84% to 94% of volunteers.

Oximony, the natural extract of sustainably sourced and FairWild certified *Lysimachia christinae*, promotes skin longevity by combating cellular senescence and supporting taurine synthesis, leading to a powerful reduction of skin fatigue in just 14 days, an improved skin firmness and glow, thereby meeting the consumer quest for longevity and healthy aging.

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and regulatory product information
- Composition data sheet
- Specifications
- Product Purity Profile
- Formulation Data Sheet
- Botanical Sourcing information
- Natural content (according ISO 16128)

Annex 2 - Clinical test formula

Trade name	INCI name	Placebo formulation %	Oximony formulation %
Glycerine 99-5	Glycerin	5.00	5.00
1,3-Butanediol	Butylene Glycol	10.00	10.00
Verdessence Xanthan	Xanthan Gum	2.00	2.00
Sodium Benzoate	Sodium Benzoate	0.25	0.25
Dermosoft 1388 Eco (Evonik)	Glycerin, Aqua, Sodium Levulinate, Sodium Anisate	1.00	1.00
Dehymuls® PGPH	Polyglyceryl-2 Dipolyhydroxystearate	1.00	1.00
Cetiol CC	Dicaprylyl Carbonate	6.00	6.00
Cetiol RLF	Caprylyl Caprylate/Caprate	10.00	10.00
Plantapon LC 7	Laureth-7 Citrate	0.50	0.50
Citric Acid (50% aqueous solution)	Citric Acid, Water	0.35	0.35
Oximony BC10187	Maltodextrin (and) Lysimachia Christinae Flower/Leaf/ Stem Extract	/	0.25
Water	Water	qsf 100	qsf 100

Annex 3 - Formulation examples

Vitality granité mask (SC-FR-24-NK-0363-001)

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	87.05	
	Calcium Chloride Dihydrate	Calcium Chloride	0.50	Complexing agent
B	Glycerin	Glycerin	5.00	Humectant
	Preservative	[Preservative]	qs	Preservative
	Verdessence® Alginate	Algin	3.50	Rheology modifier
	Verdessence® Xanthan	Xanthan Gum	0.50	Rheology modifier
C	Water, demin.	Aqua	3.00	
	OXIMONY™ BC10187	LYSIMACHIA CHRISTINAE EXTRACT, MALTODEXTRIN	0.25	Active ingredient
D	Perfume	Parfum	0.20	Fragrance

Specifications	
pH value (20°C)	6.1
Viscosity (Brookfield; DV-I+; spindle TD, Helipath; 5 rpm; 20°C)	100 000 to 120 000 mPa s

EverYourth serum (SC-FR-24-NK-0364-001)

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	93.59	
	Sodium Gluconate *(Jungbunzlauer AG)*	Sodium Gluconate	0.50	Auxiliary
	Emulgade® Verde 10 MS	Polyglyceryl-10 Stearate	1.00	Emulsifier (O/W)
	Preservative	[Preservative]	qs	Preservative
	OXIMONY™ BC10187	LYSIMACHIA CHRISTINAE EXTRACT, MALTODEXTRIN	0.25	Active ingredient
B	Verdessence® Xanthan	Xanthan Gum	1.70	Stabilizer
C	Cetiol® Ultimate	Undecane, Tridecane	2.50	Emollient
D	Perfume	Parfum	0.10	Fragrance
E	Citric Acid (10% solution)	Citric Acid	0.36	pH Adjustment

Specifications	
pH value (20°C)	5.8 - 6.0
Viscosity (Brookfield; DV-I+; spindle 5; 50 rpm; 20°C)	3 000 - 4 000 mPa s

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